Learning Objective 2.3: I can describe the radiation mechanism, pattern, and typical use cases for patch, slot, and horn antennas.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate). Take $c = 3 \times 10^8$ m/s and $\eta_0 = 377\ \Omega$.

1. **Sizing a patch.** You are designing a rectangular microstrip patch to resonate at $f_r = 3.5$ GHz on Rogers RO4003C: $\varepsilon_r = 3.55$, substrate thickness $h = 0.813$ mm. The transmission-line design set is

$$W = \frac{c}{2f_r}\sqrt{\frac{2}{\varepsilon_r + 1}} \qquad \varepsilon_{\text{eff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1 + \frac{12h}{W}\right)^{-1/2}$$

$$\frac{\Delta L}{h} = 0.412\,\frac{(\varepsilon_{\text{eff}} + 0.3)(W/h + 0.264)}{(\varepsilon_{\text{eff}} - 0.258)(W/h + 0.8)} \qquad L = \frac{c}{2f_r\sqrt{\varepsilon_{\text{eff}}}} - 2\Delta L$$

(a) Find the patch width W in millimetres.

$$\frac{c}{2f_r} = \frac{3 \times 10^8}{2(3.5 \times 10^9)} = 42.86 \text{ mm}$$

$$W = 42.86\sqrt{\frac{2}{4.55}} = 42.86(0.6630) = 28.4 \text{ mm}$$

$W = \boxed{28.4 \text{ mm}}$

(b) Find the effective permittivity ε_{eff}.

$$\frac{12h}{W} = \frac{12(0.813)}{28.41} = 0.3433$$

$$\varepsilon_{\text{eff}} = 2.275 + 1.275(1.3433)^{-1/2}$$

$$= 2.275 + 1.275(0.8628) = 3.375$$

$\varepsilon_{\text{eff}} = \boxed{3.375}$

(c) Find the edge extension ΔL and the physical length L.

With $W/h = 28.41/0.813 = 34.95$,

$$\frac{\Delta L}{h} = 0.412\,\frac{(3.675)(35.21)}{(3.117)(35.75)} = 0.412(1.1614) = 0.4785$$

$$\Delta L = 0.4785(0.813) = 0.389 \text{ mm}$$

$$L = \frac{42.86}{\sqrt{3.375}} - 2(0.389) = 23.33 - 0.78 = 22.55 \text{ mm}$$

$\Delta L = \boxed{0.389 \text{ mm}}$

$L = \boxed{22.6 \text{ mm}}$

(d) The half-wavelength in the dielectric is $\lambda_d/2 = 23.3$ mm, yet you etched $L = 22.6$ mm. Explain in one or two sentences what physical effect forces the metal to be cut short, and state which way the resonant frequency moves if you etch the full 23.3 mm instead.

The fields fringe past each open edge, so the resonator is electrically longer than the copper by ΔL at each end. Resonance is set by the electrical length, so the physical metal must be shortened by $2\Delta L$ to keep the total at $\lambda_d/2$. Etching the full 23.3 mm makes the antenna electrically too long, and the resonance moves *down* in frequency — here by roughly 3%, which is several times the patch's own bandwidth, so the antenna would be badly mismatched at 3.5 GHz.

2. **The substrate trade.** A patch is a high-Q cavity, and its impedance bandwidth for VSWR $\leqslant 2$ follows the closed form

$$\mathrm{BW} \approx 3.77\,\frac{\varepsilon_r - 1}{\varepsilon_r^2}\,\frac{h}{\lambda_0}\,\frac{W}{L}$$

Continue with the 3.5 GHz design of Question 1 ($W = 28.41$ mm, $L = 22.55$ mm, $h = 0.813$ mm, $\varepsilon_r = 3.55$).

(a) Estimate the fractional bandwidth of the RO4003C design, as a percentage.

BW = 0.91%

$\lambda_0 = c/f_r = 85.7$ mm, so $h/\lambda_0 = 0.00949$.

$$\begin{aligned}\mathrm{BW} &\approx 3.77\left(\frac{2.55}{12.60}\right)(0.00949)\left(\frac{28.41}{22.55}\right)\\ &= 3.77(0.2023)(0.00949)(1.260) = 0.0091\end{aligned}$$

About 0.9% — roughly 32 MHz of usable band.

(b) Rebuild the same 3.5 GHz patch on $\varepsilon_r = 10.2$ at the same h. Its design works out to $W = 18.11$ mm, $L = 13.35$ mm. Find the new bandwidth, and the factor by which the patch *area* shrank.

BW = 0.43%

area $\boldsymbol{\nabla}\cdot$ 2.6

$$\begin{aligned}\mathrm{BW} &\approx 3.77\left(\frac{9.2}{104.0}\right)(0.00949)\left(\frac{18.11}{13.35}\right)\\ &= 3.77(0.0884)(0.00949)(1.356) = 0.0043\end{aligned}$$

Area: $28.41 \times 22.55 = 641$ mm^2 against $18.11 \times 13.35 = 242$ mm^2, a factor of 2.6 smaller. Bandwidth fell by a factor of 2.1.

(c) Read the bandwidth formula: which single substrate parameter would you change to *buy* bandwidth without changing ε_r, and name one penalty that comes with it.

Increase the thickness h: bandwidth is directly proportional to h/λ_0. The penalties are surface waves (power trapped in the substrate instead of radiated, so efficiency and pattern quality drop), a larger probe or feed inductance that is harder to match, and a thicker, heavier board. Above about $h = 0.05\lambda_0$ the closed forms above stop being trustworthy at all.

(d) You are given 8 mm × 8 mm of board area for a 3.5 GHz element and told the link needs 2% bandwidth. State whether a single patch on any of these substrates can do the job, and what you would tell the systems lead.

No. Even the low-ε_r option is far larger than 8 mm, and the only substrate that fits the footprint ($\varepsilon_r = 10.2$, at 18 × 13 mm) is still too big *and* delivers 0.43%, not 2%. Shrinking with permittivity and widening the bandwidth pull in opposite directions on this element. Tell the lead the requirement needs a different answer: a much thicker substrate, a stacked or aperture-coupled patch, or a different element entirely — not a tweak to this one.

3. **Babinet and the slot.** A narrow half-wave slot is cut in a large, thin conducting sheet and fed across its middle.

   (a) State Babinet's complementarity relation between a slot and its complementary dipole, and say in one sentence what "complementary" means physically.

   $$Z_{\text{slot}}\, Z_{\text{dipole}} = \frac{\eta_0^2}{4}$$

   The complementary structure is the one you get by exchanging metal for air and air for metal: the slot is exactly the hole left behind if the dipole were punched out of the sheet.

   (b) The complementary dipole is trimmed to resonance, where it presents 73 Ω real. Find the slot's input impedance.

   $Z_{\text{slot}} = \boxed{487\ \Omega}$

   $$\frac{\eta_0^2}{4} = \frac{(377)^2}{4} = 3.553 \times 10^4\ \Omega^2$$
   $$Z_{\text{slot}} = \frac{3.553 \times 10^4}{73} = 487\ \Omega$$

   This is the canonical "about 485 Ω" resonant-slot value.

   (c) Now take the untrimmed $\lambda/2$ dipole, $Z = 73 + j42.5\ \Omega$. Find the slot impedance and comment on the sign of its reactance.

   $Z_{\text{slot}} = \boxed{364 - j212\ \Omega}$

   $$\begin{aligned} Z_{\text{slot}} &= \frac{3.553 \times 10^4}{73 + j42.5} = \frac{3.553 \times 10^4(73 - j42.5)}{73^2 + 42.5^2} \\ &= \frac{3.553 \times 10^4}{7135}(73 - j42.5) = 4.980(73 - j42.5) \\ &= 364 - j212\ \Omega \end{aligned}$$

   The inductive dipole becomes a *capacitive* slot: inverting a complex impedance flips the sign of the reactance. Both therefore go resonant at the same length.

   (d) A horizontal slot is cut in the vertical fin of an aircraft. State the polarization of the radiated field and justify it in one sentence, then name one reason a designer chooses a slot here rather than a monopole.

   The field is *vertically* polarized. The slot's **E** field runs *across* the cut rather than along it (fields swap under Babinet), so a horizontal slot radiates like a vertical dipole. The slot is chosen because it is flush: it has no protrusion, adds no drag, and has nothing to shear off or ice up, and it can be cut into structure that is already there.

4. **An X-band horn.** A pyramidal standard-gain horn has a rectangular aperture 16.5 cm × 12.2 cm and is used at 9.4 GHz. Aperture antennas obey $G = \eta_{ap}\, 4\pi A/\lambda^2$; take the optimum-horn value $\eta_{ap} = 0.5$ unless told otherwise.

   (a) Find the gain, in dBi.

   $G = \boxed{20.9 \text{ dBi}}$

$$\lambda = \frac{3 \times 10^8}{9.4 \times 10^9} = 3.19 \text{ cm}$$
$$A = 0.165(0.122) = 0.0201 \text{ m}^2$$
$$\frac{4\pi A}{\lambda^2} = \frac{4\pi(0.0201)}{(0.0319)^2} = 248$$
$$G = 0.5(248) = 124 = 10\,\log_{10}(124) = 20.9 \text{ dBi}$$

   (b) How far away must a receiving antenna sit for this horn to be in its far field?

   $r = \boxed{2.6 \text{ m}}$

   The largest aperture dimension is the diagonal,

$$D = \sqrt{16.5^2 + 12.2^2} = 20.5 \text{ cm}$$
$$r \geqslant \frac{2D^2}{\lambda} = \frac{2(0.205)^2}{0.0319} = 2.6 \text{ m}$$

   (c) A shorter, more sharply flared horn of the same aperture has enough edge phase error to drop η_{ap} to 0.35. What gain does it deliver, and how much was lost?

   $G = \boxed{19.4 \text{ dBi}}$

   $\text{loss} = \boxed{1.6 \text{ dB}}$

$$G = 0.35(248) = 86.9 = 19.4 \text{ dBi}$$
$$\Delta G = 10\,\log_{10}\left(\frac{0.5}{0.35}\right) = 1.6 \text{ dB}$$

   The physical area is unchanged, but the horn delivers 1.6 dB less gain, all of it lost to phase error.

   (d) Explain in two or three sentences why enlarging a horn's aperture at fixed flare length eventually stops increasing its gain, and what the phrase "optimum horn" names.

   The wave leaving the flare is spherical while the aperture is flat, so the aperture edge is farther from the virtual apex than the center and its phase lags. Widening the aperture at fixed length raises $4\pi A/\lambda^2$ but steepens that quadratic phase error, so η_{ap} falls; past a point the efficiency loses faster than the area gains and the curve turns over. The *optimum horn* is that peak — the shortest horn for a given aperture whose edge phase error is still tolerable (about $\lambda/4$ in the E-plane and $3\lambda/8$ in the H-plane), which is where $\eta_{ap} \approx 0.5$ comes from.

5. **Mechanism and selection.** Answer in words. Full sentences.

   (a) A patch is mostly a flat sheet of copper over a ground plane, and a sheet of copper is a terrible radiator. Explain what actually radiates, and why the two contributions add rather than cancel in the broadside direction.

   The cavity between the patch and the ground plane holds a half-wave standing wave, so the field points from patch to ground at one open edge and from ground to patch at the other. At each open edge the field fringes out past the conductor. The vertical parts of the two fringing fields are opposite and cancel; the horizontal parts point the same way and add. Those two fringing edges behave as two slots of width W, spaced $L \approx \lambda_d/2$ apart, driven in phase — so their contributions arrive with equal path length and add straight up, giving a broadside beam.

   (b) The two radiating slots sit only about a third of a free-space wavelength apart. Say what that implies for the E-plane beamwidth, and why the patch's directivity therefore lands near 6 dBi rather than up with the horns.

   A two-element array that short cannot form a narrow beam: the array factor barely varies across the visible half space, so the E-plane pattern stays very broad. The whole radiating structure is only a fraction of a wavelength across, and directivity tracks electrical size, so a patch radiates a broad hemispherical beam worth roughly 5 to 8 dBi . Horns get their 20-plus dBi by being many wavelengths across.

   (c) Match each application to the best of patch, slot, or horn, with a one-line justification: **(i)** the gain reference in an anechoic chamber; **(ii)** a telemetry antenna on a supersonic missile skin; **(iii)** the element of a printed phased array for a 10 GHz radar demonstrator.

   (i) Horn — built to the optimum design and calibrated, so its gain is known to a few tenths of a dB, which is exactly what a reference must be. **(ii)** Slot — it is flush with the skin, so it survives the aerodynamic and thermal environment; cavity-backing makes it radiate outward only. **(iii)** Patch — planar, light, cheap, and reproducible, so hundreds of identical elements and their feed network etch in the same step.

   (d) Your team wants the phased-array demonstrator of part (c) to cover a 10% band. State the conflict this creates with the patch element and one concrete way to resolve it.

   A thin-substrate patch delivers a few percent at best, so 10% is roughly an order of magnitude beyond a plain element — and thinning the requirement is not an option if the waveform needs the band. Concrete fixes: go to a thick, low-ε_r substrate; stack a parasitic patch above the driven one; or feed through an aperture-coupled slot, which adds a second resonance and typically buys 10 to 20%. Each costs board thickness, layers, or both ; the bandwidth has to come from somewhere.

Documentation: ____